Small-parameter asymptotics for the Liouville quantum gravity dimension

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Abstract

We give a block argument for a positive, finite limit of $(d_\gamma - 2)/\gamma^{4/3}$ as $\gamma \downarrow 0$. The coefficient is characterized by the logarithmic growth rate of a Gaussian variational problem on polygonal graphs. An exact subadditivity inequality gives this rate. The upper and lower LFPP bounds use the same finite-scale variational problem, with errors that are sublinear in the number of scales. The variational coefficient is not evaluated numerically.

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1 Statement and the auxiliary constant

The planar GFF is normalized so that the logarithmic singularity of its covariance is $\log|z-w|^{-1}$. Let h be a whole-plane GFF normalized by $h_1(0) = 0$. Set $U = (-2,2)^2$ and

$$D_\varepsilon^\xi = \inf_{P:0\to 1,\ P\subset\overline{U}} \int_P e^{\xi h_\varepsilon(z)}|dz|.$$

Paths are piecewise continuously differentiable and arclength counts multiplicity. We use the established LFPP exponent relation [3, Theorem 1.5], together with the identification of the dimension exponent in [5, Corollary 1.7]:

$$\frac{\log D_\varepsilon^{\gamma/d_\gamma}}{\log \varepsilon} \xrightarrow[\varepsilon\to 0]{\mathbb{P}} \lambda(\gamma/d_\gamma), \qquad \lambda(\gamma/d_\gamma) = 1 - \frac{2+\gamma^2/2}{d_\gamma}. \tag{1.1}$$

The map $\gamma \mapsto \gamma/d_\gamma$ is continuous and strictly increasing [3, Proposition 1.7], and $d_\gamma \to 2$ as $\gamma \downarrow 0$ [3, Theorem 1.2]. Thus $\lambda(\xi)$ is defined on an interval $(0, \xi_0)$, which suffices below.

Let W be real white noise on $(0,1) \times \mathbb{R}$ with Lebesgue control measure. For continuous $f : [0,1] \to \mathbb{R}$, define

$$A_f(x,y) = \operatorname{sgn}(f(x)) \mathbf{1}_{\{\min(0,f(x)) < y < \max(0,f(x))\}}, \qquad Z_f = \sqrt{2\pi}\, W(A_f).$$

In particular,

$$\operatorname{Cov}(Z_f, Z_g) = \pi \int_0^1 (|f| + |g| - |f-g|)\, dx, \tag{1.2}$$

$$\mathbb{E}[(Z_f - Z_g)^2] = 2\pi \|f-g\|_{L^1(0,1)}. \tag{1.3}$$

For $n \geq 1$, let $\mathcal{V}_n$ be the space of continuous functions, zero at 0 and 1, that are affine on each interval of the uniform mesh 16^{-n}. Write

$$\mathcal{E}(f) = \frac{1}{2} \int_0^1 |f'(x)|^2\, dx, \qquad a_n = \mathbb{E} \sup_{f \in \mathcal{V}_n} \{Z_f - \mathcal{E}(f)\}. \tag{1.4}$$

Suprema are taken in the separable continuous version of the finite-parameter Gaussian process. Their finiteness is proved below.

Theorem 1.1. *The sequence (a_n) is subadditive and*

$$0 < a_* := \lim_{n\to\infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n} < \infty.$$

Moreover,

$$\lim_{\xi \downarrow 0} \frac{\lambda(\xi)}{\xi^{4/3}} = \frac{a_*}{\log 16}, \tag{1.5}$$

and hence

$$\lim_{\gamma \downarrow 0} \frac{d_\gamma - 2}{\gamma^{4/3}} = \frac{2^{-1/3} a_*}{\log 16}. \tag{1.6}$$

The variational formula (1.4) *and the infimum defining a_* characterize the coefficient. They do not give a closed-form or numerical evaluation of it.*

The proof uses two finite-block comparisons. The parameter n in these comparisons is fixed before $\xi \downarrow 0$. These comparisons are proved in Sections 4 and 5; they are not additional hypotheses. At every fixed n, ξ, the LFPP regularization scale tends to zero first.

Proposition 1.2. *There is a constant $C < \infty$, independent of n, such that, for all sufficiently large integers n and $M = 16^n$,*

$$\limsup_{\xi \downarrow 0} \frac{\lambda(\xi)}{\xi^{4/3}} \leq \frac{a_n + Cn^{3/4}}{\log(2M/3)}, \tag{1.7}$$

$$\liminf_{\xi \downarrow 0} \frac{\lambda(\xi)}{\xi^{4/3}} \geq \frac{a_n - Cn^{7/8}}{\log M}. \tag{1.8}$$

Throughout the proof, C denotes a finite positive constant that may change from line to line. A subscript permits dependence on that parameter. Constants without subscripts are independent of n, ξ, and the regularization scale. The little-oh notation $o_n(1)$ refers to $\xi \downarrow 0$ with n fixed; uniformity in other arguments is stated explicitly.

2 The Gaussian variational problem

We use the following standard Gaussian facts. If two centered, separable Gaussian families have ordered canonical squared distances, then their expected suprema, with the same deterministic additive function on the index set, have the same ordering; see [2, Theorem 1.2]. To specify the deterministic-drift version, take finite families X_i, Y_i, a common drift b_i, and independent copies of X, Y. For

$$F_\beta(z) = \beta^{-1} \log \sum_i e^{\beta z_i}, \qquad Z_s = \sqrt{1-s}\, X + \sqrt{s}\, Y + b,$$

Gaussian integration by parts gives

$$\frac{d}{ds}\mathbb{E}F_\beta(Z_s) = \frac{\beta}{4}\mathbb{E}\sum_{i,j} p_i(Z_s)p_j(Z_s)\big(d_Y(i,j)^2 - d_X(i,j)^2\big),$$

where $p_i(z) = e^{\beta z_i}/\sum_j e^{\beta z_j}$. The derivative is nonnegative when $d_X \le d_Y$. Letting $\beta \to \infty$ proves the comparison with drift. Finite subsets and compact energy truncations give the separable versions used below.

If F is a supremum of affine Gaussian functions with variance parameter at most v^2, then

$$\log \mathbb{E}e^{t(F-\mathbb{E}F)} \le \frac{1}{2}t^2v^2, \qquad t \in \mathbb{R}. \tag{2.1}$$

This assertion concerns a Gaussian Lipschitz functional, not a Gaussian random variable. For a finite supremum $F(x) = \max_i(\langle v_i, x\rangle + b_i)$ of affine functions of a standard Gaussian vector, $\sup_i \|v_i\| \le v$ makes F v-Lipschitz. The Gaussian logarithmic Sobolev inequality gives, for $\psi(t) = \log \mathbb{E}e^{tF}$,

$$t\psi'(t) - \psi(t) \le \frac{t^2v^2}{2}.$$

Integrating $(\psi(t)/t)' \le v^2/2$ from zero proves (2.1) for $t > 0$; apply the same argument to $-F$ for $t < 0$. Smooth approximation and then finite-subset approximation prove the stated version. We also use the Gaussian entropy integral bound [4] and the corresponding Gaussian tail inequality [1].

Lemma 2.1. *The numbers a_n are finite and satisfy $a_{n+m} \le a_n + a_m$. In particular $a_n \le na_1$. Also,*

$$\mathbb{E}\sup_{\substack{f\in\mathcal{V}_n\\ \mathcal{E}(f)\le R}} Z_f \le Cn^{3/4}R^{1/4}, \qquad R > 0, \tag{2.2}$$

and

$$\sup_{\mathcal{E}(f)\le R} \operatorname{Var} Z_f \le C\sqrt{R}. \tag{2.3}$$

Proof. For a fixed n, the unit energy ball in $\mathcal{V}_n$ is compact in its finite-dimensional parameter space. Equation (1.3) gives a finite Gaussian entropy integral on this ball. Its expected supremum is therefore finite. Vertical dilation of the white noise gives, jointly over f,

$$(Z_{rf})_f \overset{d}{=} \sqrt{r}\,(Z_f)_f, \qquad r > 0.$$

Consequently the expected supremum on the energy-R ball is $R^{1/4}$ times that on the unit energy ball. Since $|f(x)| \le \|f'\|_2$, (2.3) follows from (1.2). Gaussian concentration on successive energy annuli now proves integrability of $\sup_f(Z_f - \mathcal{E}(f))$.

For subadditivity, write $f = g + u$, where g is the linear interpolation of $f \in \mathcal{V}_{n+m}$ on the coarser mesh 16^{-n}. Then $g \in \mathcal{V}_n$, u vanishes at every coarser mesh point, and

$$\mathcal{E}(f) = \mathcal{E}(g) + \mathcal{E}(u).$$

For two such decompositions,

$$\|g + u - g' - u'\|_1 \le \|g - g'\|_1 + \|u - u'\|_1.$$

Gaussian comparison, with the deterministic function $-\mathcal{E}(g) - \mathcal{E}(u)$, bounds the expected supremum for Z_{g+u} by that for the sum of independent copies of Z_g and Z_u. The latter expected supremum is $a_n + a_m$. Indeed, the fine problem splits over the coarser intervals. On an interval of length r, the substitution $u(x) = r\widetilde{u}((x - x_0)/r)$ multiplies both the energy and the Gaussian field in law by r. The interval contributions therefore sum to a_m.

For $b > 0$, vertical dilation also gives

$$\mathbb{E} \sup_{f \in \mathcal{V}_n} (Z_f - b\mathcal{E}(f)) = b^{-1/3} a_n.$$

Thus, writing $w_n = \mathbb{E} \sup_{\mathcal{E}(f) \le 1} Z_f$, we have $w_n \le b^{-1/3} a_n + b$. Minimizing in b, and using $a_n \le n a_1$, gives $w_n \le C n^{3/4}$. Dilation gives (2.2). □

Lemma 2.2. *For every $n \ge 1$,*

$$a_n \ge 3\, 2^{-13/3} n. \tag{2.4}$$

Proof. For each 16-adic interval $I = [l, l + r]$ at a level $j \in \{0, \dots, n-1\}$, let

$$\psi_I(x) = r\psi((x - l)/r), \qquad \psi(t) = \min(t, 1 - t) \quad (0 \le t \le 1),$$

with the function zero outside its interval. Fix $b > 0$. For sign assignments $\sigma_I \in \{-1, 1\}$, set $f_\sigma = b \sum_I \sigma_I \psi_I$. The tent derivatives are orthogonal: a coarser derivative is constant on the support of each finer tent, whose derivative integrates to zero. Hence $f_\sigma \in \mathcal{V}_n$ and $\mathcal{E}(f_\sigma) = n b^2/2$.

For two assignments, let $\mathcal{D}$ be the intervals at which their signs first differ along the ancestry. These intervals are disjoint. On the central half of an interval $I \in \mathcal{D}$ of length r, the leading difference has magnitude at least $br/2$. The sum of all finer differences has magnitude at most $br/15$. Therefore

$$\|f_\sigma - f_\tau\|_1 \ge \frac{b}{6} \sum_{I \in \mathcal{D}} |I|^2.$$

Let $G_{I,w}$ be independent standard normal variables, indexed by an interval and a word of signs along its ancestry, including the interval itself. Define

$$Y_\sigma = \sqrt{\frac{\pi b}{8}} \sum_I |I| G_{I, \sigma|I}.$$

The squared canonical distance of this family is at most

$$\frac{\pi b}{4} \frac{16}{15} \sum_{I \in \mathcal{D}} |I|^2 \le \frac{\pi b}{3} \sum_{I \in \mathcal{D}} |I|^2 \le \mathbb{E}[(Z_{f_\sigma} - Z_{f_\tau})^2].$$

A greedy sign choice, proceeding from the root to the leaves and taking the larger of the two new standard normals at each interval, has expected Y-value $n\sqrt{b}/\sqrt{8}$. Here $\mathbb{E}\max(G_1, G_2) = 1/\sqrt{\pi}$ and the interval lengths sum to one at each level. Gaussian comparison gives

$$a_n \geq n\left(\frac{\sqrt{b}}{\sqrt{8}} - \frac{b^2}{2}\right).$$

Taking $b = 2^{-5/3}$ proves the claim. □

Lemma 2.3. *For all sufficiently large n, with $M = 16^n$, there is a finite set $\mathcal{F}_n \subset \mathcal{V}_n$, containing zero, such that*

$$\mathcal{E}(f) \leq Cn, \quad \|f\|_\infty \leq C\sqrt{n} \quad (f \in \mathcal{F}_n), \tag{2.5}$$

$$\log|\mathcal{F}_n| \leq CM\log M, \tag{2.6}$$

$$\mathbb{E}\max_{f\in\mathcal{F}_n}(Z_f - \mathcal{E}(f)) \geq a_n - 1. \tag{2.7}$$

Proof. By (2.2), the expected supremum of Z on the energy-R ball is at most $Cn^{3/4}R^{1/4}$. For $R \geq Kn$ and a sufficiently large universal K, this is at most $R/4$. Equation (2.3) and Gaussian concentration then show that the expected positive contribution of the annuli $2^jKn < \mathcal{E}(f) \leq 2^{j+1}Kn$ to $Z_f - \mathcal{E}(f)$ is summable and tends to zero as $n \to \infty$. Hence restricting to $\mathcal{E}(f) \leq Kn$ changes a_n by less than $1/4$ for all large n.

Round the values at the $M-1$ interior mesh points to multiples of $\eta = M^{-3}$, leaving the endpoints zero. The uniform error is at most η, the energy of the difference is at most $CM^2\eta^2$, and the energy increase is at most $C\sqrt{n}M\eta + CM^2\eta^2$. The rounded family has cardinality at most $(C\sqrt{n}/\eta)^M$ and energy at most $2Kn$ for large n.

For completeness, within any rounding cell the canonical diameter of the Gaussian differences from its center is at most $C\sqrt{\eta}$, and a canonical u-cover has cardinality at most $(C\eta/u^2)^M$. The entropy integral bounds the mean supremum within a cell by $C\sqrt{M\eta}$. Taking the maximum over the cells adds at most $C\sqrt{\eta M\log M}$, by Gaussian concentration. Both quantities tend to zero. This proves (2.7), after increasing the energy constant. □

3 Gaussian estimates for thin polygons

For a segment or polygonal path P, let ν_P be its normalized arclength measure. All measures in pairings with h below have finite logarithmic energy. Pairings with signed measures of total mass zero do not depend on the normalization of h.

Lemma 3.1. *Suppose $R \geq r > 0$. Let μ_R, ν_R be probability measures satisfying*

$$\mu_R(B(z,t)), \nu_R(B(z,t)) \leq L_R\min(1, t/R),$$

and suppose they can be coupled with displacement at most u. Make the analogous assumptions at scale r, with constant $L_r \geq 1$ and displacement v. Set $\sigma_R = \mu_R - \nu_R$ and $\sigma_r = \mu_r - \nu_r$. Then, for $\varepsilon \geq 0$,

$$|\operatorname{Cov}(\langle h_\varepsilon, \sigma_R\rangle, \langle h_\varepsilon, \sigma_r\rangle)| \leq CL_RL_r\frac{\sqrt{uv}}{R}. \tag{3.1}$$

Here $h_0 = h$. In particular, the variance for one pair is at most CL_R^2u/R. Also, simultaneous convolution of any family of zero-mass measures with a probability measure decreases its Gaussian covariance in the positive-semidefinite order.

Proof. Let $g_t(z) = (2\pi t^2)^{-1} e^{-|z|^2/(2t^2)}$. The logarithmic covariance, with the constant term canceled by zero masses, gives

$$\mathrm{Cov}(\langle h, \sigma_R\rangle, \langle h, \sigma_r\rangle) = 4\pi \int_0^\infty t \langle \sigma_R * g_t, \sigma_r * g_t\rangle_{L^2(\mathbb{R}^2)}\, dt.$$

The growth and displacement assumptions give

$$\|\sigma_R * g_t\|_\infty \le \frac{CL_R}{Rt} \min(1, u/t), \qquad \|\sigma_r * g_t\|_1 \le C \min(1, v/t).$$

The first estimate follows by integrating the Gaussian derivative bound against the coupling when $t \ge u$, and by the growth bound otherwise. The second is the L^1 translation estimate for a Gaussian density. The covariance is therefore bounded by

$$\frac{CL_R}{R} \int_0^\infty \min(1, u/t) \min(1, v/t)\, dt \le \frac{CL_R}{R} \sqrt{uv}.$$

Indeed, the integral equals $m(2 + \log(M'/m)) \le 2\sqrt{uv}$, where $m = \min(u, v)$ and $M' = \max(u, v)$. Smooth approximation justifies the heat identity for the stated measures: the growth bound makes the logarithmic singularity uniformly integrable. Circle convolution preserves the growth and coupling bounds, proving the regularized statement.

For the last assertion, the GFF covariance norm on zero-mass measures is translation invariant. The squared norm of an average of translates is at most the average of their squared norms. Apply this to every finite linear combination of measures. □

For a continuous f define the horizontal-coordinate measure

$$\mu_{\delta,f} = \int_0^1 \delta_{(x,\delta f(x))}\, dx.$$

It is not, in general, normalized arclength.

Lemma 3.2. *Fix a finite-dimensional space of piecewise affine functions on a fixed mesh. On compact parameter sets, jointly and in the topology of uniform convergence,*

$$\delta^{-1/2} \langle h, \mu_{\delta,f+g} - \mu_{\delta,g}\rangle \xrightarrow[\delta\downarrow 0]{\text{law}} Z_{f+g} - Z_g. \tag{3.2}$$

Here f vanishes at the two endpoints, and g may range over a compact set of affine functions. Expected suprema, also after adding a uniformly convergent continuous deterministic function, converge as well.

The same limit holds if each graph and its reference segment are subjected to a common longitudinal affine displacement of order δ, and if the measures are replaced by normalized arclength. It also holds for the following equivalent parametrization: the first $M-1$ edges of the polygon have length R/M, the last edge joins the endpoint of the parent chord of length R, and the vertical vertex displacements from that chord are $\delta R f(i/M)$. On bounded parameter sets,

$$\delta^{-2} \log(\mathrm{len}(P)/R) \longrightarrow \mathcal{E}(f). \tag{3.3}$$

Proof. For $g = 0$, direct integration of the logarithmic kernel gives

$$\mathrm{Cov}\left(\delta^{-1/2}\langle h, \mu_{\delta,f} - \mu_{\delta,0}\rangle, \delta^{-1/2}\langle h, \mu_{\delta,f'} - \mu_{\delta,0}\rangle\right)$$

$$= \frac{1}{2\delta}\int_0^1\int_0^1 \log\frac{((x-y)^2+\delta^2 f(x)^2)((x-y)^2+\delta^2 f'(y)^2)}{(x-y)^2((x-y)^2+\delta^2(f(x)-f'(y))^2)}\,dy\,dx.$$

Substitute $y = x + \delta u$ and use

$$\int_{\mathbb{R}} \log(1+b^2/u^2)\,du = 2\pi|b|.$$

The limit is (1.2). The same calculation with four arbitrary graph functions gives the covariance of $Z_{f+g} - Z_g$. Bounded functions give integrable logarithmic majorants, and translated logarithmic singularities are uniformly integrable. A longitudinal affine displacement changes u by a bounded translation in all four terms; the limiting integral is unchanged.

Graph measures have the growth bound at scale one because their horizontal projections are Lebesgue measure. Lemma 3.1 gives a squared canonical distance at most $C\delta$ times the distance between the vertex parameters. After division by δ, the entropy bound on a fixed finite-dimensional compact set is uniform. This proves tightness, uniform integrability of suprema, and the stated process convergence.

For normalized arclength, the density correction relative to horizontal coordinate is $O(\delta^2)$ on these compact sets. Its Gaussian covariance norm is $O(\delta^2)$, since the finitely many segment measures have bounded logarithmic energies. Moving a vertex by $O(\delta^2)$ changes the Gaussian covariance norm by $O(\delta)$, by Lemma 3.1; these errors vanish after division by $\sqrt{\delta}$. For uniformity, the error process has pointwise covariance norm at most $C_n\delta$ and canonical modulus at most $C_n\sqrt{\delta\|p-q\|}$ in the vertex parameters. Its entropy integral is at most $C_n\delta\sqrt{1+\log(1/\delta)} = o_n(\sqrt{\delta})$. The arclength density error is smaller.

Finally, in coordinates along the parent chord, write $f_i = f(i/M)$. The horizontal increment of each of the first $M-1$ edges is

$$\frac{1}{M}\sqrt{1-\delta^2M^2(f_i-f_{i-1})^2} = \frac{1}{M} - \frac{\delta^2 M}{2}(f_i-f_{i-1})^2 + O(\delta^4).$$

The last edge joins the endpoint. Expanding its length yields total relative length $1+\frac{1}{2}\delta^2 M\sum_{i=1}^{M}(f_i - f_{i-1})^2 + O(\delta^4)$. The coefficient is $\mathcal{E}(f)$. The remainders are uniform on the fixed compact parameter sets. Small endpoint perturbations add an affine function g; the cross term in the energy is zero because $\int_0^1 f' = 0$. □

Lemma 3.3. *Let $M = 16^n$, $\eta = M^{-2}$, and let $\mathcal{A}_\eta$ be the affine functions with both endpoint values bounded in absolute value by $C_0\eta$, for a fixed constant C_0. For $k \geq 0$,*

$$\mathbb{E}\sup_{\substack{f\in\mathcal{V}_n,\ \mathcal{E}(f)\leq k+1\\ g\in\mathcal{A}_\eta}}\{Z_{f+g} - Z_g - k\} \leq \min\{a_n + C,\ Cn^{3/4}(k+1)^{1/4} - k + C\}. \tag{3.4}$$

Proof. For fixed g, the process $Z_{f+g} - Z_g$ has the law of Z_f. For fixed f, changing g to g' has squared canonical distance at most $C\|g-g'\|_1$, since each of the two interval endpoints is shifted by $g - g'$. Thus, for $s > 0$, the squared canonical distance of the two-parameter process is bounded by the sum of $(1+s)$ times the squared canonical distance of Z_f and $C(1+s^{-1})$ times that of an independent copy indexed by g.

Gaussian comparison and dilation give

$$\mathbb{E}\sup_{f,g}\{Z_{f+g} - Z_g - \mathcal{E}(f)\} \leq (1+s)^{2/3}a_n + C\sqrt{\eta(1+s^{-1})}.$$

Take $s = 1/n$ and use $a_n \leq Cn$. On $\mathcal{E}(f) \leq k+1$, replacing $-\mathcal{E}(f)$ by $-k$ increases the expression by at most one. The second bound follows from the same comparison and (2.2). □

Lemma 3.4. *There is a white noise $\widehat{W}$ on $\mathbb{R}^2 \times (0,\infty)$ with control measure $dy\,dt$ such that, for*

$$K_t(z) = \pi^{-1/2} t^{-3/2} e^{-|z|^2/(2t^2)}, \qquad G_{a,b}(z) = \int_a^b \int_{\mathbb{R}^2} K_t(z-y)\widehat{W}(\,dy,\,dt),$$

the following hold. Disjoint scale intervals are independent, the fields have exact similarity scaling, and

$$\mathrm{Cov}(G_{a,b}(z), G_{a,b}(w)) = \int_a^b e^{-|z-w|^2/(4t^2)} \frac{dt}{t}. \tag{3.5}$$

In particular, the point variance is $\log(b/a)$. *These fields can be coupled with the normalized whole-plane GFF so that, for every compact V and fixed $\rho > 0$,*

$$\sup_{z\in V} |h_\varepsilon(z) - G_{\varepsilon,\rho}(z)| = O_{\mathbb{P}}(\sqrt{\log(1/\varepsilon)}) \quad (\varepsilon \downarrow 0). \tag{3.6}$$

Also, for a fixed compact V,

$$\langle h_\varepsilon, \nu_{[0,1]} \rangle = O_{\mathbb{P}}(1), \qquad \sup_{\substack{z,w\in V \\ |z-w|\le 8\varepsilon}} |h_\varepsilon(z) - h_\varepsilon(w)| = O_{\mathbb{P}}(\sqrt{\log(1/\varepsilon)}). \tag{3.7}$$

Proof. Convolution of two kernels gives (3.5). For zero-mass test measures, integrating this covariance over $(0,\infty)$ gives the logarithmic covariance. Subtraction of the uniform unit-circle measure fixes the additive constant. Similarity scaling follows from the factor $r^{3/2}$ in the scaling of white noise, which cancels the factor $r^{-3/2}$ in the kernel.

Here are the estimates for (3.6). If σ_ε is the uniform circle measure of radius ε, then

$$\|\sigma_\varepsilon * K_t\|_2^2 \le C/\varepsilon \quad (t \le \varepsilon), \qquad \|\sigma_\varepsilon * K_t - K_t\|_2^2 \le C\varepsilon^2/t^3 \quad (t \ge \varepsilon).$$

The first follows by integrating the Gaussian overlap kernel on two copies of the circle, and the second from the L^2 derivative bound. Integrating in t bounds the point variance of the difference uniformly in ε. The circle normalization and scales above ρ add only a bounded variance on V. For $|z-w| \le \varepsilon$, Lemma 3.1 bounds the circle-average increment variance by $C|z-w|/\varepsilon$, whereas the smooth field increment variance is at most $C|z-w|^2/\varepsilon^2$. A canonical u-cover therefore has size at most $C_V \varepsilon^{-2} u^{-4}$. The Gaussian entropy integral and concentration prove (3.6).

The first assertion in (3.7) follows from the bounded logarithmic energies of the line and circle measures. For the second, the circle-average increment family has bounded variance and a canonical u-cover of size at most $C_V \varepsilon^{-2} u^{-8}$. Apply the same Gaussian bounds. □

4 Upper bound for the LFPP exponent

Fix $n \ge 1$, put $M = 16^n$, and set

$$\delta = \xi^{2/3}, \qquad \eta = M^{-2}.$$

The auxiliary constants in estimates made before letting $\delta \downarrow 0$ may depend on n. The constants in the limiting estimates below do not.

4.1 A partition and its deterministic inequality

For a path P admitted in the definition of D_ε^ξ, recursively partition its consecutive subpaths as follows. A node u has chord endpoints x_u, y_u, length $R_u > 0$, and $P_u \subset \overline{B}(x_u, 4R_u)$. The root has endpoints $0, 1$. If $R_u > \varepsilon$, cut at successive first displacements of length $r = R_u/M$. Retain a terminal residual piece if its chord is longer than $r/2$, and otherwise merge it with the preceding complete piece. All children then satisfy

$$\frac{R_u}{2M} \le R_v \le \frac{3R_u}{2M}, \tag{4.1}$$

and the same subpath ball condition. All child chords except possibly the last have length R_u/M. Stop when $R_u \le \varepsilon$.

The resulting tree is finite. Indeed, a rectifiable subpath has only finitely many complete first-exit pieces, and chord lengths decrease by a uniform factor. Put

$$S_u = \sum_{v \text{ child of } u} R_v, \quad A_u = \log(S_u/R_u), \quad q_{uv} = R_v/S_u.$$

Let θ be the unit flow with transition weights q_{uv}, and write

$$H_u = \langle h_\varepsilon, \nu_{[x_u, y_u]} \rangle, \qquad \Delta_u = \sum_v q_{uv} H_v - H_u.$$

If Q_u is the polygon of child chords, then $\Delta_u = \langle h_\varepsilon, \nu_{Q_u} - \nu_{[x_u, y_u]} \rangle$. The leaf estimates and Jensen's inequality give

$$\log \int_P e^{\xi h_\varepsilon} |dz| \ge \xi H_{\text{root}} + \sum_{u \text{ internal}} \theta(u)(A_u + \xi \Delta_u) - \xi \omega_\varepsilon, \tag{4.2}$$

where ω_ε is the oscillation in (3.7). To verify the identity used here, for a leaf v one has

$$R_v = \theta(v) \exp\Big(\sum_{u \prec v} A_u \Big), \qquad \sum_{v \text{ leaf}} \theta(v) H_v = H_{\text{root}} + \sum_{u \text{ internal}} \theta(u) \Delta_u.$$

A leaf path has cost at least $R_v e^{\xi H_v - \xi \omega_\varepsilon}$. Summing and applying Jensen proves (4.2).

Put $b_u = A_u \wedge 1$ and $k_u = \lfloor b_u/\delta^2 \rfloor$. If $A_u \le 1$, set $G_u = -\Delta_u$. If $A_u > 1$, choose an adverse child maximizing $H_u - H_v$ and set $G_u = H_u - H_v$. Then

$$-(A_u + \xi \Delta_u) \le \xi G_u - b_u \le \delta^2 (\delta^{-1/2} G_u - k_u). \tag{4.3}$$

The last equality of scales uses $\xi = \delta^{3/2}$.

4.2 Records and Gaussian bounds

Assign a chord the dyadic scale L such that $R \in (L/2, L]$, and use a square grid of side $\eta \delta L$. A chain record specifies the current scale and parent endpoint cells, its bin k, whether $A \le 1$, the child followed by the chain as described below, and the next scale and endpoint cells. For $A \le 1$, record the followed child's chronological index. For $A > 1$, no chronological index is recorded; only its endpoint cells are needed. The adverse chord in the latter case will be included in the local Gaussian supremum and need not be recorded.

Lemma 4.1. *There are universal C, D with the following properties.*

Given a parent's scale and endpoint cells, the number of next records with bin k is at most

$$CM^D(k+1)^D. \tag{4.4}$$

Let $m_{\delta,n}(k)$ be a common upper bound for the mean of the local supremum of $\delta^{-1/2}G - k$ in any such record, with the endpoint and polygon parameters allowed to vary continuously. One can take

$$m_{\delta,n}(k) \le C_n(k+1)^{1/4} - k, \tag{4.5}$$

and, for each fixed k,

$$\limsup_{\delta\downarrow 0} m_{\delta,n}(k) \le \min\{a_n + C,\ Cn^{3/4}(k+1)^{1/4} - k + C\}. \tag{4.6}$$

For a chain record of length l and bins $k_1, \dots, k_l$, uniformly over choices of its Gaussian variables,

$$\operatorname{Var}\Big(\sum_{i=1}^{l} \delta^{-1/2} G_i\Big) \le C \sum_{i=1}^{l} g_{\delta,n}(k_i)^2 \sqrt{k_i + 1}, \tag{4.7}$$

where $1 \le g_{\delta,n}(k) \le CM$ and, for every fixed k, $g_{\delta,n}(k) \to 1$ as $\delta \downarrow 0$. Increasing the universal constant in (4.7) *is understood.*

Proof. If $A \le 1$, the number of edges is at most CM. Writing e for the direction of the parent chord and t_j for the unit child directions gives

$$\sum_j R_j |t_j - e|^2 = 2(S - R).$$

Since all edges before the last have length R/M, every interior vertex z_i satisfies

$$|z_i - x - (i/M)(y - x)| \le CR\sqrt{A}.$$

For bin k this is at most $C\delta R\sqrt{k+1}$. The next grid has spacing comparable to $\eta\delta R/M$. There are CM possible indices and at most a fixed power of $M\sqrt{k+1}/\eta$ possible endpoint-cell pairs. Since $\eta = M^{-2}$, this gives (4.4). If $A > 1$, the followed endpoints lie in $B(x, 4R)$. Their number of possible cell pairs is at most $CM^D\delta^{-4}$, which is bounded by $CM^D(k+1)^2$ because $k = \lfloor \delta^{-2} \rfloor$. There are only a universal number of choices for the next dyadic scale.

For $A \le 1$, normalized arclength on the polygon and chord can be coupled within $CR\sqrt{A}$. This follows by normalized-arclength parametrization and

$$\int_0^1 |Q'(t) - (y - x)|^2 \, dt = S^2 - R^2 \le CAR^2.$$

Their growth constants at scale R are at most CM. If $A \le c/M$ for a sufficiently small universal c, the preceding direction identity makes every child point forwards with angle bounded away from $\pi/2$. The polygon is then a graph of uniformly bounded slope, and its growth constant is universal. In the large-excess case the two chord measures have growth constants at most CM and can be coupled within CR.

Thus we may take $g_{\delta,n}(k) = 1$ if $\delta^2(k+1) \le c/M$, and $g_{\delta,n}(k) = CM$ otherwise. Lemma 3.1 gives the covariance bound

$$Cg_{\delta,n}(k_i)g_{\delta,n}(k_j)\sqrt{L_j/L_i}\,(k_i+1)^{1/4}(k_j+1)^{1/4} \quad (i < j)$$

for the normalized variables. The assigned scales along a record decrease by at least $M/4$ at every step. The resulting geometric matrix has uniformly bounded row sums for $M \geq 16$. Applying $2uv \leq u^2 + v^2$ proves (4.7). The same estimate holds for the regularized field by Lemma 3.1.

Here are details of the mean bounds. For fixed n, the small-excess polygons have a bounded number of vertex parameters, all within distance $C\delta R\sqrt{k+1}$ of the corresponding straight positions; the parent endpoint uncertainty is smaller. Moving the parameters by τR changes the Gaussian variable by at most $C_n\sqrt{\tau}$ in covariance norm. The variance is at most $C_n\delta\sqrt{k+1}$. Covering these parameter boxes and applying the Gaussian entropy integral gives a mean supremum at most $C_n\delta^{1/2}(k+1)^{1/4}$. For the large-excess case, the unnormalized mean supremum over the two chord measures is at most C_n; here $(k+1)^{1/4} \geq \delta^{-1/2}$. These statements prove (4.5), uniformly over all record locations and scales. The centered zero-mass field law is invariant under similarities. For each record, normalize by the segment joining the centers of its two endpoint cells. Its length is comparable to the assigned scale, and the normalized endpoint uncertainty is bounded by $C\eta\delta$. Consequently the bounds are uniform over record positions, orientations, and the ratio of chord length to assigned dyadic scale.

For fixed k and $\delta \downarrow 0$, only the small-excess case remains. There are exactly M edges for all small δ: the first $m-1$ have length R/M, the last has length in $[R/(2M), 3R/(2M)]$, and $S/R \to 1$, which forces $m = M$. Lemma 3.2 identifies the limiting local family with

$$\{Z_{f+g} - Z_g - k : \ f \in \mathcal{V}_n, \ \mathcal{E}(f) \leq k+1, \ g \in \mathcal{A}_{C\eta}\}.$$

The parent endpoint cells produce the affine function g; longitudinal cell uncertainty has no effect on the limit. The compact-parameter convergence and uniform integrability in Lemma 3.2, followed by Lemma 3.3, prove (4.6). Taking the full field instead of h_ε only enlarges these mean suprema, by Gaussian comparison and the last assertion of Lemma 3.1. The sets of indices in this comparison may depend on δ: uniform process convergence applies on a fixed compact parameter set, and every limiting admissible profile has energy at most $k+1$. No conditioning on the occurrence of a record is imposed. □

4.3 Summation over chains

For $t > 0$ define

$$\mathcal{Z}_{\delta,n}(t) = CM^D \sum_{k=0}^{\infty} (k+1)^D \exp\left(t m_{\delta,n}(k) + Ct^2 g_{\delta,n}(k)^2\sqrt{k+1}\right). \tag{4.8}$$

Bins that do not occur can be omitted; including them with the bounds above only enlarges the series. It is finite by (4.5). For a fixed chain record, enlarge its configuration space to the product of its local spaces. The supremum of the sum then equals the sum of the local suprema, regardless of their dependence. Its mean is at most $\sum_i m_{\delta,n}(k_i)$, and its variance parameter is bounded by (4.7). Equation (2.1) and the record count therefore imply

$$\mathbb{P}\left[\text{some length-}l \text{ chain has } \sum_i (\delta^{-1/2}G_i - k_i) > B_{\delta,n}(t)l\right] \leq e^{-l},$$

where

$$B_{\delta,n}(t) = \frac{\log \mathcal{Z}_{\delta,n}(t) + 1}{t}.$$

A root-to-leaf chain has length between $\log(1/\varepsilon)/\log(2M)$ and $\log(1/\varepsilon)/\log(2M/3) + 1$. Summing the last probability over these lengths gives a quantity tending to zero as $\varepsilon \downarrow 0$. Averaging the

simultaneous chain bound against the leaf flow in (4.2), and using (4.3) and (3.7), gives

$$\lambda(\xi) \le \delta^2 \frac{B_{\delta,n}(t)}{\log(2M/3)}. \tag{4.9}$$

For fixed n, t, the summands in (4.8) are dominated uniformly in δ by a summable sequence, using (4.5) and $g_{\delta,n} \le CM$. Hence (4.6) gives

$$\limsup_{\delta\downarrow 0} \log \mathcal{Z}_{\delta,n}(t) \le \log(CM^D) + \log \sum_{k\ge 0} (k+1)^D e^{t\min(a_n+C,\, Cn^{3/4}(k+1)^{1/4}-k+C)+Ct^2\sqrt{k+1}}.$$

Take $t = n^{1/4}$. For $k \le Kn$, the logarithm of this sum is at most $ta_n + Cn$, after choosing a sufficiently large fixed K. For $k > Kn$, use $Cn^{3/4}(k+1)^{1/4} \le k/4$ and absorb $Ct^2\sqrt{k+1}$ into $tk/4$; the remaining tail is summable and no larger than the preceding bound. Since $\log M = n \log 16$, we obtain

$$\limsup_{\delta\downarrow 0} B_{\delta,n}(n^{1/4}) \le a_n + Cn^{3/4}.$$

Equation (4.9) proves (1.7).

5 Lower bound for the LFPP exponent

We retain the block size n, $M = 16^n$, and $\delta = \xi^{2/3}$. Let $\mathcal{F}_n$ be the family in Lemma 2.3. Choose $K_n \le C\sqrt{n}$ so that $\|f\|_\infty \le K_n$ for every $f \in \mathcal{F}_n$, and put

$$R_n = n^{3/4}, \qquad \rho = \delta R_n, \qquad X = G_{4\rho/M,\rho}, \qquad v = \log(M/4).$$

Multiplying R_n by a fixed constant, or restricting to larger n, does not change the estimates. We take n sufficiently large that the relative errors below are less than $1/2$, and then take ξ sufficiently small for this fixed n.

For $f \in \mathcal{F}_n$, let P_f have vertices $(i/M, \delta f(i/M))$, $0 \le i \le M$. Its ordered edge lengths are

$$r_i^f = \frac{1}{M}\sqrt{1+\delta^2 p_i(f)^2}, \qquad p_i(f) = M(f(i/M) - f((i-1)/M)).$$

Thus $1/M \le r_i^f \le 2/M$ for all small δ, uniformly over the finite family. Let T_i^f be the orientation-preserving similarity from $[0,1]$ onto edge i, and write $\ell_f = \sum_i r_i^f$. Then

$$\ell_f = 1 + \delta^2 \mathcal{E}(f) + O_n(\delta^4). \tag{5.1}$$

Set

$$T_\delta = \{z : \operatorname{dist}(z, [0,1]) \le 2\delta K_n\}.$$

Every recursive replacement by these polygons stays in T_δ, by summing the displacements $\delta K_n (2/M)^j$ along an ancestry.

5.1 A bound uniform over all refinement measures

For an arbitrary deterministic probability measure μ supported on T_δ, define

$$C_f(\mu) = \sum_{i=1}^{M} r_i^f \int e^{\xi X(T_i^f z)} \mu(\,dz), \qquad \nu_\mu^f = \frac{1}{M} \sum_{i=1}^{M} (T_i^f)_* \mu. \tag{5.2}$$

The measures ν_μ^f are deliberately weighted by $1/M$, rather than by edge lengths.

Lemma 5.1. *There is a universal C such that*

$$\sup_{\mu \in \mathcal{P}(T_\delta)} \mathbb{E} \min_{f \in \mathcal{F}_n} C_f(\mu) \le 1 - \delta^2 (a_n - C n^{7/8}) + o_n(\delta^2). \tag{5.3}$$

The remainder is uniform over the probability measures μ, including singular measures.

Proof. Define the centered Gaussian families

$$Y_f = \delta^{-1/2} \langle X, \nu_\mu^f - \nu_\mu^0 \rangle, \qquad U_f = \delta^{-1/2} \langle h, \mu_{\delta,f} - \mu_{\delta,0} \rangle,$$

using the coupling in Lemma 3.4. Let

$$d(f,g)^2 = 2\pi \|f - g\|_1.$$

We first prove, uniformly in μ, that for all sufficiently small δ,

$$\operatorname{Var}(Y_f - Y_g) \ge (1 - C n^{-1/4}) d(f,g)^2 - C \frac{n^{3/4}}{M} \qquad (f, g \in \mathcal{F}_n). \tag{5.4}$$

Here is the potential calculation that supplies the uniformity. For fixed f, g, define

$$\Phi_\delta^{f,g}(x,y) = \frac{1}{\delta} \int_0^1 \log \frac{|(x, \delta y) - (s, \delta g(s))|}{|(x, \delta y) - (s, \delta f(s))|} \, ds.$$

Changing variables $s = x + \delta u$ gives

$$\Phi_\delta^{f,g}(x,y) \longrightarrow \pi(|y - g(x)| - |y - f(x)|). \tag{5.5}$$

The convergence is uniform for bounded y and $x \in [0,1]$. It remains uniform if x extends by $O(\delta)$ beyond the endpoints, with f, g extended by zero there. To see this directly, the transformed integrand is

$$\frac{1}{2} \log \frac{u^2 + (y - g(x + \delta u))^2}{u^2 + (y - f(x + \delta u))^2}.$$

For $|u| > A$ with A larger than a fixed bound for the profile and y values, its absolute value is at most C_n / u^2. On $|u| \le A$, the profile arguments change by at most $C_n \delta A$. Uniform integrability near $u = 0$ follows by bounding the negative part of each logarithm by $2|\log |u||$; away from zero uniform continuity applies. If x is within δA of an endpoint, the error from completing the constant-profile integral to the whole real line is at most $2\pi |f(x) - g(x)|$, which tends uniformly to zero. The same argument applies just outside the endpoints, with zero extension. This proves the asserted uniform convergence.

Write $z = u + iv \in T_\delta$, $x_i = (i - 1 + u)/M$, and $\zeta = v/(\delta M)$. For $u \in [0,1]$, up to a longitudinal error $O_n(\delta^2)$, the two points $T_i^f z, T_i^g z$ have scaled coordinates

$$(x_i, f(x_i) + \zeta), \qquad (x_i, g(x_i) + \zeta), \qquad |\zeta| \le 2K_n / M.$$

Applying (5.5) to their difference gives

$$\pi\big(|w(x_i)+\zeta|+|w(x_i)-\zeta|-2|\zeta|\big)=2\pi(|w(x_i)|-|\zeta|)_+, \qquad w=f-g.$$

This is at least $2\pi|w(x_i)|-CK_n/M$. For every $u\in[0,1]$,

$$\left|\frac{1}{M}\sum_{i=1}^{M}|w((i-1+u)/M)|-\int_0^1|w(x)|\,dx\right|\le\frac{\mathrm{TV}(|w|)}{M}\le\frac{\|w'\|_1}{M}.$$

When u lies outside $[0,1]$ by $O_n(\delta)$, the affine extension of edge i differs from the neighboring piece of the graph by $O_n(\delta)$ in scaled vertical coordinate. Uniformity in (5.5) shows that this overshoot adds only $o_n(1)$. Since $\|w'\|_1\le C\sqrt{n}$, the full logarithmic kernel would give a cross covariance at least

$$d(f,g)^2-C\sqrt{n}/M+o_n(1).$$

Every remainder here is uniform over μ because the pointwise bounds are uniform on its support.

We next remove the scales outside the band defining X. For scales $t<4\rho/M$, each point-to-line kernel is bounded by

$$\int_0^{4\rho/M}\frac{dt}{t}\int_0^1 e^{-(x-s)^2/(4t^2)}\,ds\le C\rho/M.$$

After division by δ, removing these scales changes the cross covariance by at most CR_n/M. For scales $t>\rho$, use a mixed second derivative of the Gaussian kernel for the point difference and the reference-line difference. The point displacement is $\delta|w(x_i)|+O_n(\delta^2)$, and the reference displacement is at most $2\delta K_n$. The Gaussian kernel satisfies

$$\|D_zD_{z'}e^{-|z-z'|^2/(4t^2)}\|\le Ct^{-2}e^{-|z-z'|^2/(8t^2)}.$$

The reference points retain horizontal coordinate s, so integrating this bound in $s\in[0,1]$ gives $Ct^{-2}\min(t,1)$, uniformly over the intermediate points used in the mixed difference. Since

$$\int_\rho^\infty t^{-3}\min(t,1)\,dt\le C/\rho\quad(\rho<1),$$

the absolute contribution, after division by δ, is at most $C(K_n/R_n)|w(x_i)|+o_n(1)$. Averaging and using the preceding quadrature estimate proves

$$\mathrm{Cov}(Y_f-Y_g,U_f-U_g)\ge(1-Cn^{-1/4})d(f,g)^2-C\frac{n^{3/4}}{M}+o_n(1),\tag{5.6}$$

uniformly in μ. This computation involves only deterministic kernel potentials; no pointwise value of the unregularized GFF is used.

Here is an explicit absorption of the remainders in this step. Increase fixed universal constants and set

$$\alpha_n=C_1n^{-1/4},\qquad\beta_n=C_2n^{3/4}/M.$$

Take n large enough that $\alpha_n<1/2$. By Lemma 3.2, the variance of U_f-U_g converges to $r=d(f,g)^2>0$ for every distinct pair in the finite family. The remainder in (5.6) is uniform in μ. Consequently, by decreasing a threshold $\delta_0(n)>0$, for every distinct pair and every μ we have

$$\mathrm{Cov}(Y_f-Y_g,U_f-U_g)\ge(1-\alpha_n)r-2\beta_n,\qquad\mathrm{Var}(U_f-U_g)\le(1+\alpha_n)r.$$

Cauchy–Schwarz and the inequality $(Ar-\beta)_+^2/r \geq A^2 r - 2A\beta$ imply

$$\operatorname{Var}(Y_f - Y_g) \geq \frac{\big((1-\alpha_n)r - 2\beta_n\big)_+^2}{(1+\alpha_n)r} \geq (1-3\alpha_n)r - 4\beta_n.$$

For $f = g$ the desired lower bound is immediate. This proves (5.4). Small separations between members of the finite family affect only $\delta_0(n)$, not the universal constants or uniformity in μ.

Add independent centered Gaussian variables N_f, each with variance $Cn^{3/4}/M$, to the family Y_f. By (5.4), their variance can be chosen so that the canonical squared distances of $Y + N$ dominate those of $s_n Z$, where $s_n \geq 1 - Cn^{-1/4}$. Gaussian comparison with the deterministic function $-\mathcal{E}(f)$, (2.5), and (2.7) imply

$$\begin{aligned}
\mathbb{E}\max_{f\in\mathcal{F}_n}\{-Y_f - \mathcal{E}(f)\} &\geq \mathbb{E}\max_{f\in\mathcal{F}_n}\{-s_n Z_f - \mathcal{E}(f)\} - \mathbb{E}\max_f(-N_f) \\
&\geq a_n - Cn^{3/4} - C\sqrt{\frac{n^{3/4}}{M}\log|\mathcal{F}_n|} \\
&\geq a_n - Cn^{7/8}.
\end{aligned}$$

In the second line, use $\mathcal{E}(f) \leq Cn$, $a_n \leq Cn$, and $s_n \geq 1 - Cn^{-1/4}$. The estimate holds uniformly in μ for all small δ.

Finally, Taylor expansion in (5.2), together with (5.1), gives

$$C_f(\mu) = 1 + \delta^2\mathcal{E}(f) + \xi\langle X, \nu_\mu^f\rangle + \mathcal{R}_f.$$

Uniformly over μ, $\mathbb{E}\max_{f\in\mathcal{F}_n}|\mathcal{R}_f| \leq C_n(\xi^2 + \delta^4 + \xi\delta^2) = o_n(\delta^2)$. Indeed, X has fixed point variance $v = \log(M/4)$, the family is finite, and integration against probability measures preserves the pointwise Gaussian moment bounds. The difference between the edge weights r_i^f and $1/M$ in the linear term is included in this remainder. The common centered Gaussian variable $\xi\langle X, \nu_\mu^0\rangle$ has mean zero and can be removed after taking the minimum. Since $\xi/\delta^2 = \delta^{-1/2}$, the last maximum bound proves (5.3). □

5.2 Iteration with unequal child lengths

We now iterate the block choice. This step accounts explicitly for all unused scales. Fix n and ξ as above, and let

$$\varepsilon_j = \rho M^{-j}, \qquad H^{(j)} = G_{\varepsilon_j,\rho}.$$

At depth zero, take $P_0 = [0,1]$, $m_0 = 1$, and μ_0 uniform on that segment. Suppose the depth-$(j-1)$ rule and its expected weighted occupation measure have been defined. Set

$$M_{j-1}(A) = \mathbb{E}\int_{P_{j-1}} \mathbf{1}_A(z)e^{\xi H^{(j-1)}(z)}|dz|, \qquad m_{j-1} = M_{j-1}(\mathbb{C}), \qquad \mu_{j-1} = M_{j-1}/m_{j-1}.$$

The measure μ_{j-1} is deterministic: it is formed from an unconditional expected weighted occupation measure, not the realized occupation measure or a conditional one. Using only the root band X, select a minimizer of $C_f(\mu_{j-1})$ over $\mathcal{F}_n$, with a fixed rule for ties. In every selected child apply a similarity-transformed copy of the depth-$(j-1)$ rule, using the finer white noise. This defines P_j. There are finitely many possible polygons at every finite depth, so the rule and the occupation measures are well-defined. The support of every μ_j is contained in T_δ.

Condition on the entire root noise band. The child similarities, of scales $r_i \in [1/M, 2/M]$, are then fixed. The child rule uses noise with upper scale $\rho r_i < 4\rho/M$ and natural lower cutoff $\varepsilon_{j-1} r_i$. Under inverse similarity, the white noise below ρr_i has the original law and is independent of the conditioned root band. Its conditional expected occupation measure, weighted by $G_{\varepsilon_{j-1} r_i, \rho r_i}$, is

$$r_i m_{j-1}(T_i)_* \mu_{j-1}.$$

The two remaining scale intervals for this child's contribution are

$$(\rho r_i, 4\rho/M), \qquad (\varepsilon_j, \varepsilon_{j-1} r_i).$$

Neither interval is used in choosing the child path. The first lies above its entire driving noise. The second lies below its natural cutoff; all bands used by a depth-$(j-1)$ child rule have lower endpoints at least $4\varepsilon_{j-1} r_i$, since every child ratio is at least $1/M$. Conditional on the root noise, these two unused intervals are independent of both the child path and its weighted field $G_{\varepsilon_{j-1} r_i, \rho r_i}$. This assertion is needed separately for each child; it does not assert independence from the driving noise of every other child. Their point variances sum to exactly

$$\log \frac{4}{M r_i} + \log(M r_i) = \log 4.$$

Integrating their exponential weights therefore gives $4^{\xi^2/2}$, independently of the point and the child. Possible correlations between different children do not matter, since only their individual conditional expectations are added. We obtain the exact scalar recursion

$$m_j = 4^{\xi^2/2} m_{j-1} \mathbb{E} \min_{f \in \mathcal{F}_n} C_f(\mu_{j-1}). \tag{5.7}$$

The two unused intervals also account for the case $j = 1$, when the depth-zero child carries no field.

Lemma 5.1 is uniform in the refinement measure. Thus, for fixed n,

$$m_j \le b_{\xi,n}^j, \qquad b_{\xi,n} = 4^{\xi^2/2} \left[1 - \delta^2 (a_n - C n^{7/8}) + o_n(\delta^2)\right].$$

For all sufficiently large n and then sufficiently small ξ, this factor is positive and less than one. Markov's inequality shows that, for each fixed $s > 0$,

$$\mathbb{P}\left[\int_{P_j} e^{\xi H^{(j)}(z)} |dz| > e^{j(\log b_{\xi,n} + s)}\right] \le e^{-js}.$$

The paths lie in $T_\delta \subset U$. Equation (3.6) changes their logarithmic cost by $o_{\mathbb{P}}(j)$ when $H^{(j)}$ is replaced by h_{ε_j}. Comparing with (1.1) and letting $s \downarrow 0$ yields

$$\lambda(\xi) \ge -\frac{\log b_{\xi,n}}{\log M}.$$

Since $\xi^2 = o(\delta^2)$, this proves (1.8).

6 Conclusion

Proof of Theorem 1.1. Lemmas 2.1 and 2.2, followed by Fekete's lemma, give

$$3\,2^{-13/3} \le a_* := \lim_{n\to\infty} a_n/n = \inf_{n \ge 1} a_n/n \le a_1 < \infty.$$

Let $n \to \infty$ in the two bounds of Proposition 1.2. Since $\log M = n \log 16$, both right-hand sides tend to $a_*/\log 16$. This proves (1.5).

For $\xi_\gamma = \gamma/d_\gamma$, the exact relation (1.1) gives

$$\frac{d_\gamma - 2}{\gamma^{4/3}} = \frac{1}{2}\gamma^{2/3} + d_\gamma^{-1/3}\frac{\lambda(\xi_\gamma)}{\xi_\gamma^{4/3}}.$$

Since $d_\gamma \to 2$, equation (1.6) follows. □

Remark 6.1. The unregularized Gaussian functional cannot be minimized over all finite-energy paths: its infimum is $-\infty$, even on a countable bounded-energy family. The constant in Theorem 1.1 is instead the growth rate of the cutoff problems (1.4). For example, in the tent construction in Lemma 2.2, taking the coefficient $b = n^{-1/2}$ keeps $\int |f'|^2 = 1$ while the expected Gaussian maximum is at least $n^{3/4}/\sqrt{8}$. Its individual variances are $O(n^{-1/2})$, so Gaussian concentration and Borel–Cantelli give the asserted divergence on the union of these finite families.

Remark 6.2. The displayed lower bound on a_* gives the nonoptimized bound

$$\frac{2^{-1/3}a_*}{\log 16} \geq \frac{3\,2^{-14/3}}{\log 16}.$$

This is a lower bound, not an evaluation of the limiting coefficient. No claim of an independently verified numerical value is made.

References

[1] C. Borell. The Brunn–Minkowski inequality in Gauss space. *Invent. Math.* **30** (1975), 207–216.

[2] S. Chatterjee. An error bound in the Sudakov–Fernique inequality. arXiv:math/0510424, 2005.

[3] J. Ding and E. Gwynne. The fractal dimension of Liouville quantum gravity: universality, monotonicity, and bounds. *Comm. Math. Phys.* **374** (2020), 1877–1934. arXiv:1807.01072.

[4] R. M. Dudley. The sizes of compact subsets of Hilbert space and continuity of Gaussian processes. *J. Funct. Anal.* **1** (1967), 290–330.

[5] E. Gwynne and J. Pfeffer. KPZ formulas for the Liouville quantum gravity metric. arXiv:1905.11790, version 3.